

KATHIRVEL

Full Stack Developer · Java · FastAPI · React.js · LangChain · AI/RAG

📍 Madurai, Tamil Nadu 📞 +91-6381111323 ✉ kathirvelv1r108@gmail.com 🌐 GitHub 🌐 LinkedIn 🌐 Portfolio

Professional Summary

Results-driven Full Stack Developer and Software Developer Intern with hands-on experience building scalable web applications and AI-powered systems using Java, Spring Boot, FastAPI, React.js, and Node.js. Specialized in LangChain-based **RAG pipelines**, semantic search, vector databases, and LLM integration. Proven ability to ship secure REST APIs with JWT authentication and RBAC, and deliver end-to-end projects from design to deployment. Passionate about generative AI and intelligent application development.

Technical Skills

Languages	Java, Python, JavaScript (ES6+), SQL
Frontend	React.js, HTML5, CSS3, Responsive Web Design, Component-Based Architecture
Backend	Spring Boot, FastAPI, Node.js, Express.js, REST API Design, JWT Auth, RBAC, Microservices
AI / LLM	LangChain, RAG Pipelines, Prompt Engineering, Semantic Search, Embedding Workflows, Vector Databases, Ollama
Databases	MySQL, MongoDB, ChromaDB
DevOps / Cloud	Docker, Git, GitHub, CI/CD, AWS EC2, AWS S3, Linux
Concepts	OOP, DSA, DBMS, SDLC, Agile, API Security, System Design Fundamentals

Experience

Software Developer Intern | Spiritmine Tech

Mar 2026 – Present Tamil Nadu

- Engineered scalable full-stack applications using **React.js**, **Node.js**, and RESTful APIs in Agile sprint cycles, contributing to 3+ production feature releases.
- Architected **JWT-based authentication** and **role-based access control (RBAC)** systems, securing multi-role access across admin and user modules.
- Built and optimized **AI-powered RAG pipelines** using **LangChain**, ChromaDB, and semantic search to enable intelligent document querying with contextual response generation.
- Developed production-grade backend APIs using **FastAPI**; participated in **Docker**-based deployment workflows, reducing environment inconsistency issues.

Projects

AI-Based RAG Document Assistant

React.js · Node.js · LangChain · ChromaDB · Ollama · FastAPI

- Designed a full-stack **Retrieval-Augmented Generation (RAG)** application enabling natural-language querying of multi-format (PDF/DOCX) documents with contextual response generation.
- Built **embedding and retrieval pipeline** using LangChain + ChromaDB + Ollama (local LLM), achieving accurate semantic search with sub-second retrieval on 500+ page documents.
- Developed scalable Node.js backend APIs with document chunking, embedding generation, and **streaming LLM inference** for real-time responses; React.js frontend with chat UI and file upload.

SmartPark – Parking Management System

Spring Boot · React.js · MySQL · JWT · REST API

- Built RESTful parking slot booking platform with **Spring Boot**, delivering CRUD operations, slot availability tracking, and workflow management for Admin and User roles.
- Implemented **JWT authentication** and **RBAC**, enforcing granular permissions; designed normalized **MySQL** schema with transaction management and concurrent booking support.
- Built a responsive **React.js** frontend featuring real-time slot status visualization, praised for its intuitive design and seamless end-to-end functionality.

Education

B.E. Computer Science and Engineering | Latha Mathavan Engineering College – Anna University

2021 – 2025

Tamil Nadu

Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Software Engineering, OOP, Computer Networks

Certifications & Achievements

- Java Full Stack Developer Certification** – QSpiders Training Institute (Java, React.js, REST APIs, MySQL)
- Solved **60+ DSA problems** across Arrays, Strings, Trees, Hashing, and Recursion on LeetCode.